THERMOREGULATION

PHYSIOLOGY DEPARTMENT- CHU CONSTANTINE

THERMOREGULATION

→1) Definition

→ 2) Mechanism of heat exchange

→ 3) Thermal regulation

→ 4) Temperature measurement

→ 5) Pathophysiology

1)- DEFINITIONS

HEAT :

1° Waste of cellular metabolism produced continuously.

2° Optimal temperature for chemical/enzymatic reactions (37°C)

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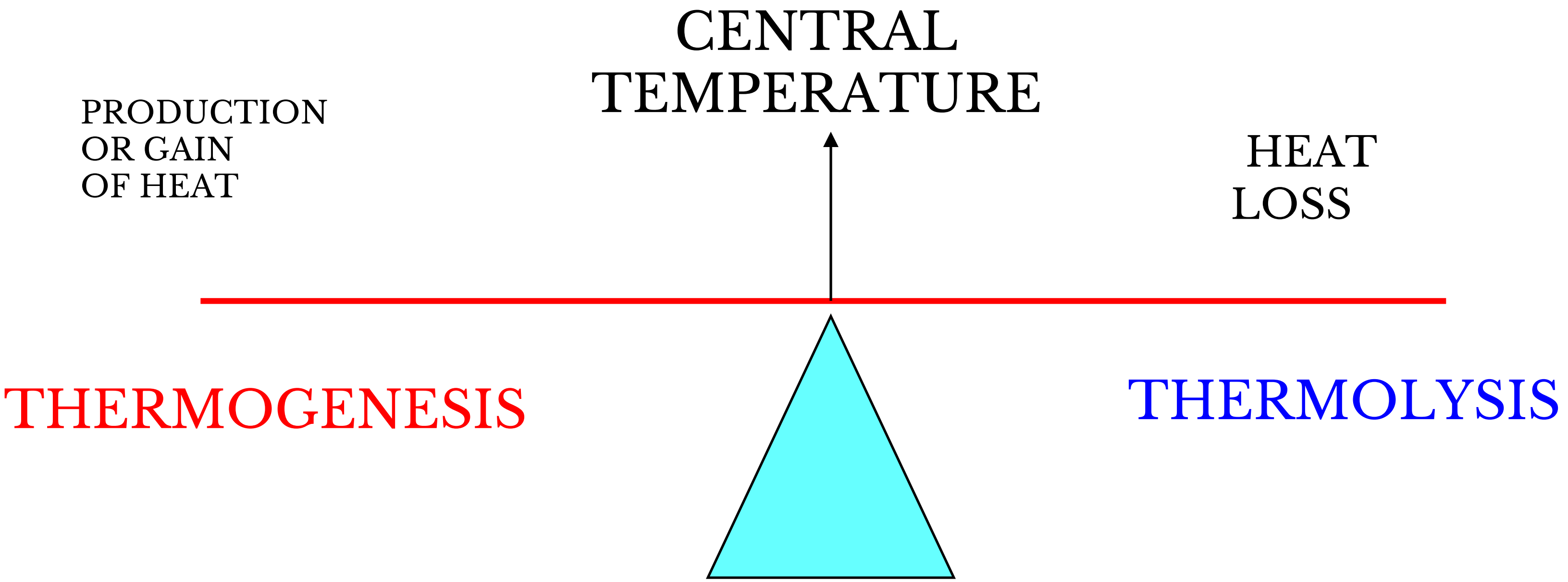
HOMEOTHERMS	HIBERNANT	POIKILOTHERMS
Central Température independant Of ambient Environment by internal heat production	H I B E R N A N T	Central temperature Dependant on the ambient environment

1) DEFINITION



Body temperature results from the balance between heat production and heat loss.

All tissues produce heat.

At rest: mainly liver, heart, brain, inactive skeletal muscles (20-30%)

EXERCISE: heat produced by skeletal muscles = 30-40 times heat produced by the rest of the body.

Body temperature (Tcorp) ranges between 36.1 and 37.8°C regardless of external temperature or heat produced by the body.

Tcorp : minimum = morning(07h00-09h00) ; maximum = evening (17h00-19h00).

- if Tcorp ↑ rate of chemical reactions ↑↑
- if Tcorp ↑↑↑ , denaturation of enzymatic and structural proteins occurs, and neuron activity in the nervous system decreases drastically ↓↓

2). Mechanisms of heat exchange

- Physical mechanisms governing heat exchange between our skin and the external environment are the same as those regulating heat transfer between inanimate objects.
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- Heat moves along its concentration gradient: **from hotter to colder**.

  Four mechanisms of heat transfer:

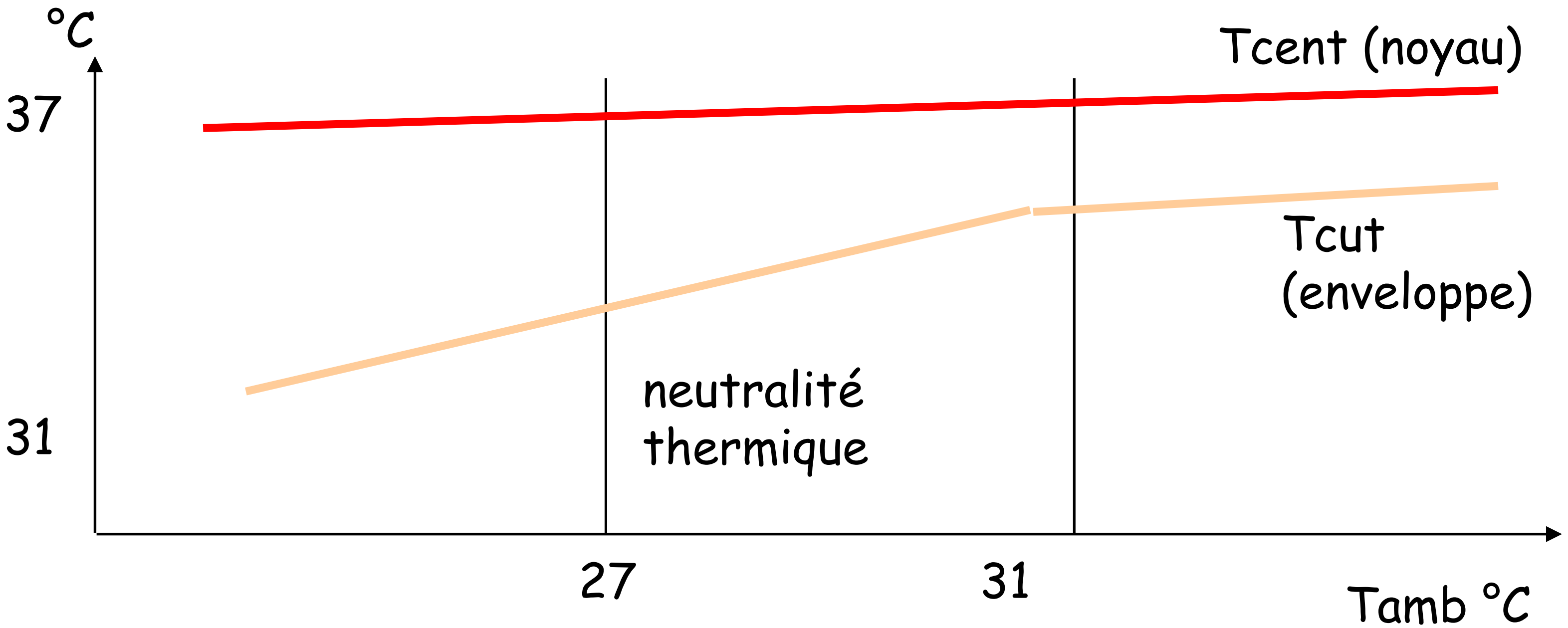
- **Radiation – Conduction – Convection - Evaporation**

PHYSICAL BASIS OF HEAT EXCHANGE

- RADIATION: heat loss (or gain) in the form of infrared waves.
- 50% of heat loss is by radiation.
- CONVECTION: cooling of the skin by wind (fan) or water.

- CONDUCTION : heat transfer between two objects in direct contact (e.g., hot bath transfers heat by conduction to the skin)
- EVAPORATION
  - **Insensible losses (respiration)**
  - **600 mL/24h = 390 Kcal/24h**
  - **Losses via urine, feces**
  - **SWEATING +++**
  - **1 mL water = 0.58 kcal**

3) THERMAL REGULATION



-> Circadian and seasonal rhythm **minimum at 5h et maximum at 20 h**

Role of the hypothalamus

Hypothalamus

- **Main integration center for thermoregulation : acts as thermostat**
- Anterior part defends against heat **(thermolysis)**
- Posterior part defends against cold **(thermogenesis)**

Role of blood and circulation

- **blood = transfer agent of heat between thee body's interior and surface.**

1) If surface hotter than outside ⇒ **heat loss** ⇒ warm blood flows in skin capillaries (**vasodilatation**)
2) If surface cooler than outside ⇒ **heat must be conserved** ⇒ blood largely bypasses subcutaneous capillary network (**vasoconstriction**) ⇒ limits heat loss

cent temperature is fairly stable; peripheral temperature can fluctuate widely(20 to 40°C)

A. Mechanisms of thermogenesis

Thermogenesis = heat production

If external temperature or blood T ↓: hypothalamic thermogenesis center is activated.

Triggers mechanisms to maintain or increase central body temperatue

1- Vasoconstriction of skin blood vessels

Activation of sympathitic nervous fibers

- Stimulation of smooth muscles in skin arterioles fff
- **Vasoconstriction**
- **Blood restricted to deeper regions , diverted from subcutaneous capillaries**

- **reduced heat loss** (hypodermis acts as an inslator)
  - Prolonged vasoconstriction may cause tissue necrosis

2- increased metabolic rate

A-Stimulation of sympathic nerves

- Release of noradrénaline
- ↑ increased metabolism of target cells
- increased glycogen use (increased O2 consumption ↑)

↑ more heat = chimical thermogenesis

B- increased thyroxine release

- If external temperature drops

⬾ hypothalamus activated … TRH released (thyréolibérine) … anterior pituitary activated , TSH secrered (thyréostimuline)

⬾ thyroid stimulated … thyroxine released into blood

↑increased metabolism in target cells

□ ↑ increased heat roduction

3- Shivering

- If previous mechanisms fail to restore temperature

⇐ Activation of brain centers regulating muscle tone

⇐ involuntary contraction of skeletal muscles = shivering

□ ↑ heat production = ↑ T corp … increased body temperature

Human adaptations :

- Wearing warm clothes to prevent heat loss
- Drinking warm liquids
- Increasing muscle activity
- Changing posture to reduce exposed body surface (crossing arms)

B. Mechanisms of thermolysis

Thermolysis = heat dissipation

- various mechanisms for heat loss : radiation - conduction - convection – évaporation

T cent > normal :

- hypothalamic thermogenesis center = inhibited
- **hypothalamique thermolysis center = activated**

this causes serval reactions

1- Vasodilatation of skin arterioles

- Modulation of sympathetic nerve fibers
- Stimulation of smooth muscles in skin arterioles decreases ↓

⇦ **Vasodilatation**

- Warm blood flows into skin vessels
- heat dissipates from skin surface via : radiation , conduction and convection

2- increased sweating

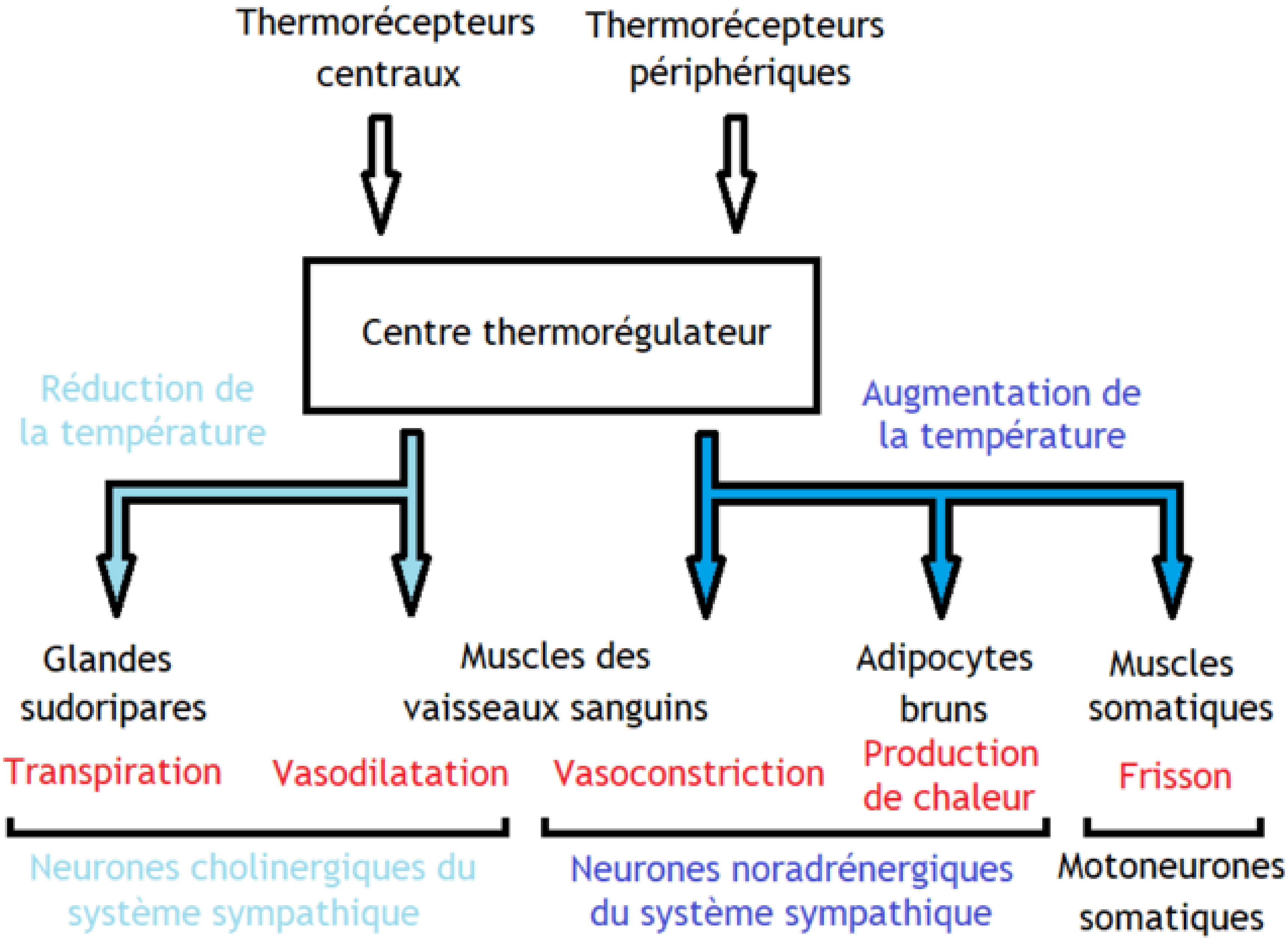
$\uparrow\uparrow$ external temperature

- Stimulation of sympathic nerves
- **Stimulation of sweat glands :** $\uparrow$ sweat production
- sweat evaporation causes **heat loss**
  - **If humidity > 60% : sweating is impaired**

    heat loss becomes difficult

Human adaptations

- Wearing loose, light-colored clothing to reduce heat gain
- Being dressed is cooler than nude because bare skin absorbs solar radiation
- Seeking cool environments
- Increasing convection (fans)
- Decreasing external temperature (air conditioning)


Thermorécepteurs centraux
Thermorécepteurs périphériques
Centre thermorégulateur
Réduction de la température
Augmentation de la température
Glandes sudoripares
Muscles des vaisseaux sanguins
Adipocytes bruns
Muscles somatiques
Transpiration
Vasodilatation
Vasoconstriction
Production de chaleur
Frisson
Neurones cholinergiques du système sympathique
Neurones noradrénergiques du système sympathique
Motoneurones somatiques

physiological variation

External factors :

- Ambient temperature, climate

Physiological factors

- Ovulation period in women
- Age (elderly 35,5-36,5: feel cold easily)
- Emotional state
- Physical work

- Time of day when temperature is taken

Measurement factors

- Anatomical region
- Measurement method

Temperature measurement

- • CENTRAL temperature
- (organ temperature)
- – oral
- – Rectal
- – Esophageal
- – Tympanic

- • PERIPHERAL TEMPERATURE (LOWER)
- – average skin temperature
- – difference peripheral/central = reflets body shell

Les méthodes de prise de température

METHODE	SITE DE LA MESURE	DUREE	INTERET	INCONVENIENTS
RECTALE	La pointe du thermomètre dépasse la marge anale	1’30’’ à 2’	Température de référence	Inconfort / septique Risque de lésion anale Non respect de la pudeur
BUCCALE	La pointe est positionne en sublingual, les lèvres bien fermées	1 minute avec un thermomètre électronique	Simple non invasif	Imprécis Variations importantes Risque septique
AXILLAIRE	La pointe est placée dans le creux de l’aisselle, le coude contre le corps	3 à 5 minutes	Simple non invasif asepsie	Des Imprécisions Longueur de temps
TYMPANIQUE	L’embout est positionné dans l’axe du conduit auditif externe face au tympan	Lecture quasi-immédiate	Facile / Précis Reflet de la t° C Asepsie Urgence confort	Petit conduit auditif Contre indiqué : Otite / trauma du rocher / bouchon de cérumen

5)-PHYSIOPATHOLOGY

- a. HYPOTHERMIA

- Cent temperature < 35°C
- Decreased muscle strength
- Intense shivering
- Decreased metabolism of drugs

- T cent <34°C

- Mental confusion
- Loss of consciousness

- T cent < 28°C

- Bradycardia
- Arrhythmia
- Ventricular fibrillation

- **CAUSES :**
- extreme conditions, anesthesia, alderly, hypothyroïdism
- **TREATMENT**
- gradual warming : infusion
- clinical use : extracorporeal circulation (ECC)

b. HYPERTHERMIA

T cent > 38°C max 42°C

Loss of thermoregulation

- reduced sweating
- increased central temperature
- low blood pressure
- loss or reduction of reflexes
- convulsions and brain death (above >42°)

CAUSES

- hereditary disease
- deshydration
- hyperthyroïdism,...

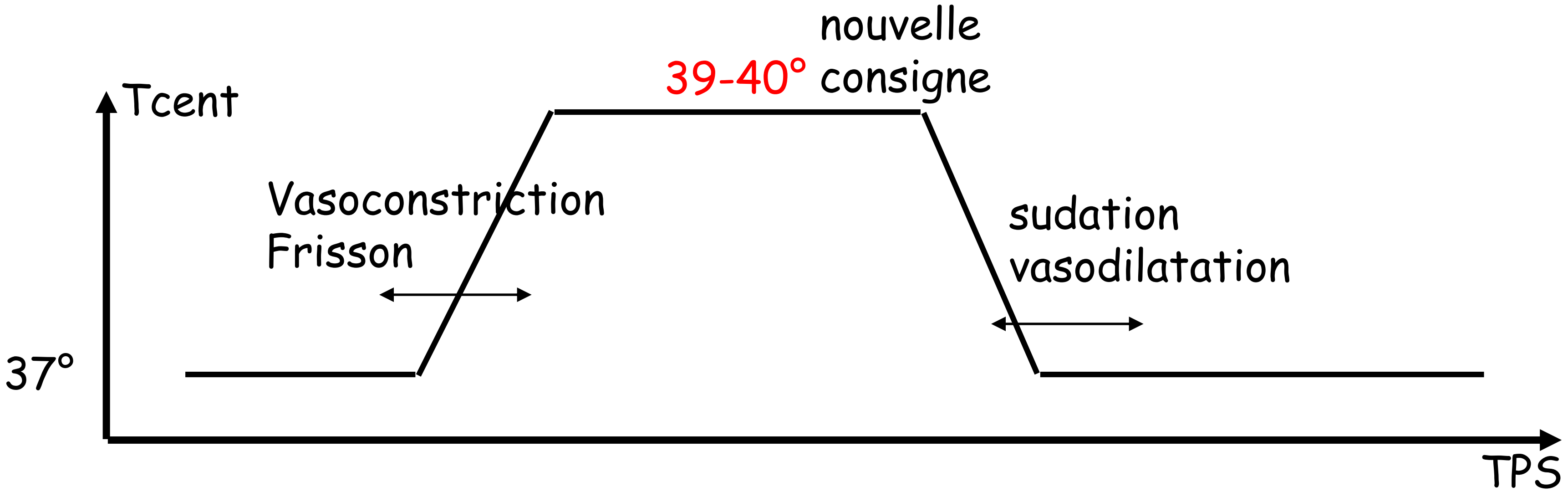
TREATMENT

- gradual cooling
- rehydration

-FEVER

- Hyperthermia due to dysregulation of regulatory mechanisms
- Infectious agents...
- Bacterial pyrogens = endotoxins
- Endogenous pyrogens = immune response to infection (e.g., IL6)

→ SET POINT SHIFT



Thermoregulation during fever

hen thermostat temperature is set higher (e.g 40°C instead of 37°), the body reacts as if in a cold environment; thermogenesis mechanisms are activated to reach the new set point: skin vasoconstriction,shivering, increased metabolism, leading to sweating.

Consequences of fever :

— increased cellular metabolism : increased heart rate and cardiac output.

— increased protein catabolism : weight loss.

— water loss : risk of dehydration.

— if temperature rises to 41 °C risk of convulsions. Especially in children with lower threshold in younger children.

CONCLUSIONS

- Thermoregulation involves no single specialized organ
- Humans are better adapted to heat
- Regulation by the central nervous system
- Clinical importance: control of hyperthermia and fever, effects of anesthetics on thermoregulation
- Adaptation (acclimatization))